A Major Project Report
(Phase-I)

On

Generative AI-Based Personalized Patient Health Information Assistant

Submitted in partial fulfillment of the requirement for the award of the degree of

BACHELOR OF TECHNOLOGY

IILM UNIVERSITY

Session 2026-27

in

Computer Science and Engineering

By

Priyanshu Kumar Yadav (2410030396)

Shreesh Rathore (2410030400)

Suraj Tyagi (2410031729)

Mohd Faiz(2410031599)

Under the guidance of Mr Atul Kumar Singh

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

IILM UNIVERSITY, GREATER NOIDA

[August], 2026

CANDIDATE'S DECLARATION

We hereby certify that the work presented in the **Phase-I report** of the project entitled "**Generative AI-Based Personalized Patient Health Information Assistant**", in partial fulfillment of the requirements for the degree of **Bachelor of Technology** in the **School of Computing Science and Engineering, IILM University**, Greater Noida, is an original work carried out by us under the supervision of **[Guide Name]**, School of Computing Science and Engineering, IILM University, Greater Noida.

The work presented in this Phase-I report has not been submitted by us for the award of any other degree or qualification at this or any other institution.

Shreesh Rathore (2410030400)
Priyanshu Kumar Yadav(2410030396)
Suraj Tyagi(2410031729)
Mohd Faiz (2410031599)

This is to certify that the above statement made by the candidates is correct to the best of my knowledge.

Signature of Guide Name and Details of Guide

Name: Mr Atul Kumar Singh

Designation: Project Mentor

School: School of Computing Science and Engineering

IILM University, Greater Noida

Table of Contents

SL. No	Section	Page no.
	Abstract	3
1	Problem Definition	3
2	Literature Survey	4
3	Objectives	7
4	Scope	7
5	Feasibility Study	8
	References	9
	Plagiarism Report	

ABSTRACT

Hospitals and clinics today sit on huge amounts of patient data, records and medical literature, but most patients still cannot get health information that is personalized and actually explained in a way they understand. Existing health-information systems tend to be static and one-way: they push out the same generic content to every user and rarely hold a real, context-aware conversation, and this gap between technical medical knowledge and what a patient can actually grasp hurts health literacy, engagement and how well people stick to their treatment. This Phase-I report lays out the problem definition, literature survey, objectives, scope and feasibility study behind a proposed Generative AI-Based Personalized Patient Health Information Assistant: a MERN-stack web application (MongoDB, Express.js, React.js, Node.js) that uses Google's Gemini API to answer patient health queries in a way that is personalized, context-aware and easy to follow, based on the health profile, symptoms and history the user provides. Based on a survey of generative-AI models used in healthcare — ChatGPT, Med-PaLM and BioGPT among them — this study works out the main strengths and limitations of such systems, including hallucination, data-privacy concerns and the need for human oversight, and uses these to shape the system's design, alongside a feasibility check across technical, economic, operational and schedule dimensions. The end goal is a system that improves health literacy, keeps patients more engaged and takes some routine workload off healthcare providers, while staying strictly an informational and triage-support tool under clinical supervision rather than a diagnostic system.

1. PROBLEM DEFINITION

Health portals, clinical-literature repositories and online communities have made a huge amount of medical information publicly available, yet patients still struggle to find clear,

personal and timely answers to their own health questions. Most health-information platforms hand out one-size-fits-all content that ignores a patient's actual medical history, current symptoms or health profile — the same static article gets shown to everyone regardless of their situation. On top of that, medical terminology and clinical documentation are often too dense for an average patient to follow unaided, which breeds confusion, anxiety and weaker adherence to treatment or preventive advice.

A second, connected problem is accessibility. A patient with a minor query, a symptom to clarify, or a general health question often cannot reach a clinician right away, which pushes them toward delayed care-seeking or unverified sources on the internet instead. Traditional health portals and static FAQ pages were never designed for a natural back-and-forth conversation, so they can't adjust to a patient's specific concern or handle a follow-up question the way a conversational system could. This absence of conversational, context-aware interaction adds to the administrative load already carried by clinicians, who spend a considerable part of their working time answering repetitive, low-complexity questions — time that an intelligent assistant could free up for more critical patient care.

Without a reliable, medically grounded assistant to turn to, patients often end up checking symptoms and self-educating through unverified sources, which raises the risk of misdiagnosis, inappropriate self-treatment or needless anxiety. The problem this project addresses is therefore two-fold: first, there is no accessible, conversational system that explains medical information in a personalized and easily understandable way based on a patient's own health profile; second, the generative-AI health tools patients are already starting to use informally largely lack proper safeguards — disclaimers, escalation pathways and a clear restriction to informational rather than diagnostic use. The proposed Generative AI-Based Personalized Patient Health Information Assistant tries to close this gap by pairing the Google Gemini API's natural-language capability with a MERN-stack web application that securely stores each user's health profile and chat history, so that every response is both personalized and kept within clear, safe operational limits.

2. LITERATURE SURVEY

To identify design principles and safeguards relevant to the proposed assistant, this section reviews recent literature across three areas: generative AI in consumer and clinical health information, domain-specific medical language models, and the risks of deploying large language models (LLMs) in healthcare. The key studies are summarised below.

2.1 Generative AI and Large Language Models for Consumer Health Information

Tilton, Caplan and Cole (2025) discuss how LLM-powered generative AI is reshaping consumer health by boosting health literacy and supporting personalized health education, pointing out that AI chatbots can break down complex medical terminology

and tailor responses to common health questions, which makes the information more accessible to people with limited medical background. In a related direction, a scoping review of how health consumers use and perceive information from generative-AI chatbots — built on the Arksey and O'Malley protocol and covering papers from 2019 onward — reports rising use of such tools for everyday symptom and treatment questions, though trust in the answers they give remains mixed. Sallam's (2023) widely cited review of ChatGPT's role in healthcare education, research and practice adds to this picture, laying out promising applications such as patient education and simplifying clinical information, while also flagging genuine concerns around accuracy and over-reliance. A more recent PMC-indexed study of a national health-information portal with an integrated generative-AI chatbot backs this up further: its multidimensional expert evaluation found that portal-integrated conversational assistants can meaningfully lift health-literacy outcomes when they are properly designed and assessed — which supports building a conversational assistant directly into a structured health-information platform rather than shipping it as a separate, standalone tool.

2.2 Domain-Specific and Clinically-Tuned Medical Language Models

In the foundational Med-PaLM paper published in Nature, Singhal et al. (2023) introduced MultiMedQA, a benchmark that pulls together several medical question-answering datasets, along with a human-evaluation framework that scores factuality, comprehension, reasoning, potential harm and bias; through instruction prompt tuning, Med-PaLM's answers to consumer medical questions held up well against clinician answers on this framework. Its successor, Med-PaLM 2, built on this with medical-domain fine-tuning and an ensemble-refinement prompting strategy, reaching up to 86.5% on the MedQA benchmark and, in a pairwise ranking of more than a thousand consumer medical questions, being preferred over physician answers on eight of nine clinical-utility measures by a panel of physicians. What this shows is that general-purpose LLMs can be steered — through fine-tuning and structured prompting rather than full retraining — into producing clinically credible, long-form answers to patient-style questions, which is directly relevant to this project's approach of prompting the Gemini API with health-profile context instead of training a model from scratch. Separately, Luo et al. (2022) put forward BioGPT, a domain-specific generative transformer trained on large-scale biomedical literature, which beat earlier biomedical language models on relation-extraction and question-answering tasks and showed the value of domain-adapted pre-training for fluent, biomedically grounded text. Taken together, these studies confirm that both general-purpose LLMs, when prompted and evaluated carefully, and domain-specific models are technically capable of producing medical text of practical quality for patients and researchers alike — which is why this project opts for a general-purpose, API-accessible model like Gemini for a student prototype rather than a self-hosted domain-specific model.

2.3 Risks, Limitations and Safety Considerations of LLMs in Healthcare

For all these capabilities, the literature keeps coming back to hallucination — plausible-sounding but factually wrong output — as the single biggest safety concern for medical LLM deployment. A 2025 multi-model assurance analysis found that in up to 83% of cases, leading LLMs went along with fabricated clinical details planted in adversarial prompts, and mitigation prompts cut this risk down without removing it. A JMIR perspective on harm-reduction strategies for medical LLM use puts empirical hallucination rates above 60% for open-ended clinical tasks and notes that misinformation together with privacy and data-security risk make up most of the overall risk picture for both patients and clinicians. A wider survey of the opportunities and challenges facing ChatGPT and LLMs in biomedicine reaches a similar conclusion, pointing to hallucination, limited transparency and an inability to fully capture clinical nuance as reasons these systems need careful human oversight instead of being left to run autonomously. On the privacy front, reviews of LLM-based healthcare chatbots — among them a privacy-by-design framework for clinical chatbots and a review of privacy challenges in retrieval-augmented generation for healthcare — document risks like re-identification of sensitive data, unclear data-retention practices, and users wrongly assuming consumer-facing chatbots offer healthcare-grade confidentiality. Together, these findings justify the safeguards built into the proposed system: explicit disclaimers, keeping the assistant to informational and triage support rather than diagnosis, prompting escalation to a qualified professional for concerning symptoms, and handling user health data securely within the MongoDB-backed profile store.

2.4 Research Gap

Pulling this literature together: (a) generative AI chatbots are technically able to improve health literacy and simplify medical information for a general audience; (b) domain-specific and instruction-tuned LLMs such as Med-PaLM and BioGPT can generate clinically credible, long-form medical answers; and (c) hallucination and data-privacy risks are well documented and need to be mitigated deliberately at the design level rather than left to the base model. That said, most of the systems reviewed here are either research prototypes tested on benchmark question sets, single-purpose chatbots bolted onto an existing national health portal, or domain-specific models that need substantial computing infrastructure to run. Very few combine a persistent, patient-specific health profile — symptoms, history, prior conversations — with a general-purpose, API-accessible generative model inside a lightweight, self-contained MERN application that treats safety scaffolding (disclaimers, escalation prompts, a clearly informational-only scope) as a first-class design requirement rather than something added on afterward. That gap is what this project's proposed Generative AI-Based Personalized Patient Health Information Assistant sets out to fill.

3. OBJECTIVES

This project has the following primary objectives:

- Design and build a MERN-stack web application where patients can register, log in and keep a personal health profile covering basic demographic details, known conditions, symptoms and medical history.
- Integrate the Google Gemini API into the Node.js/Express.js backend so it can process patients' natural-language queries and generate coherent, context-aware and empathetic health-information responses grounded in the user's stored health profile.
- Build an intuitive, responsive React.js chat interface that lets patients ask follow-up questions and get simplified explanations of medical terms, conditions and general health guidance.
- Store user profiles, chat history and health records securely in MongoDB, with access control that keeps sensitive health information visible only to the authenticated user.
- Build in explicit safety safeguards — disclaimers, escalation prompts that recommend professional consultation for concerning symptoms, and keeping the assistant to informational and triage support rather than diagnostic decision-making — informed directly by the risks surfaced in the literature survey.
- Assess the technical, economic, operational and schedule feasibility of building the proposed assistant as a Phase-I academic prototype.

4. SCOPE

This project is scoped as a web-based, patient-facing informational assistant, not a diagnostic or clinical-decision-support system. Within that scope, the platform will cover the following functional areas:

- User registration and a personal health-profile module that captures demographic details, known conditions, symptoms and medical history.
- A chat-style conversational interface for submitting health queries and getting personalized, easy-to-follow responses generated through the Gemini API.
- Simplified explanations of medical terms and conditions, plus general health and wellness guidance, tailored to the stored health profile and the conversation so far.
- Secure storage and retrieval of user profiles and chat history in MongoDB, backed by basic authentication and session management.
- Built-in disclaimers and escalation messages that point users toward professional consultation when symptoms go beyond general information.

At this Phase-I stage, the following are considered out of scope: autonomous diagnosis or treatment prescription, integration with hospital or clinic electronic health record (EHR) systems, real-time integration with wearable or IoT health-monitoring devices, and multilingual or voice-based interaction. These could be taken up as future extensions once

the core platform has been validated. The project is not meant to replace consultation with a qualified healthcare professional — it is meant to work as a supportive, informational layer that helps patients understand things better before, between or after clinical visits, always under appropriate human oversight.

5. FEASIBILITY STUDY

5.1 Technical Feasibility

The proposed platform can be built entirely on mature, well-documented and freely available technology: React.js for a responsive front end, Node.js and Express.js for the backend API layer, and MongoDB as a flexible document store for user profiles, health records and chat history. Natural-language understanding and generation are handed off to Google's Gemini API over HTTPS, so the project never needs to train or host its own language model. Since comparable generative-AI health-information systems — Med-PaLM, Med-PaLM 2, BioGPT and several consumer-facing chatbot deployments reviewed in Section 2 — have already been built and evaluated in the literature, both the core technology stack and the general approach of prompting a hosted LLM with user context can be considered mature and technically achievable for a project of this scope.

5.2 Economic Feasibility

As an academic Phase-I prototype, this project can run on free-tier cloud hosting, open-source frameworks (the whole MERN stack) and the free usage tier of the Gemini API, which keeps direct development cost close to zero. If the platform were ever scaled past an academic prototype, the main future cost would come from paid API usage at higher query volumes plus possible compliance costs for handling sensitive health data at scale — both of which can be introduced gradually and are not needed for the Phase-I prototype.

5.3 Operational Feasibility

Operationally, the platform mainly depends on the reliability and quality of the Gemini API's responses, and on applying the safety scaffolding identified in the literature survey — disclaimers, escalation prompts, scope restrictions — consistently across every conversation. Adoption risk is managed by keeping the chat interface simple and familiar, close to the messaging apps people already use, and by grounding every response in the user's own health profile so answers feel personal rather than generic. The literature's call for human oversight is met at this Phase-I stage through clear in-app disclaimers rather than live clinician review; that gap is acknowledged as a limitation to revisit in later phases.

5.4 Schedule Feasibility

Following the Phase-I to Phase-II structure of the Major Project course, this phase has focused on problem definition, literature survey, requirement gathering and feasibility

assessment, all completed within the time available. Design, development and testing of the actual working prototype — the health-profile module, the Gemini-API-integrated chat interface and the safety scaffolding around it — are planned for Phase-II, building on the system architecture and requirements set out in this report.

REFERENCES

- Singhal, K., Azizi, S., Tu, T., et al. (2023). Large language models encode clinical knowledge. Nature, 620, 172–180.
- Singhal, K., Tu, T., Gottweis, J., et al. (2023/2025). Towards expert-level medical question answering with large language models (Med-PaLM 2). Nature Medicine / arXiv:2305.09617.
- Luo, R., Sun, L., Xia, Y., Qin, T., Zhang, S., Poon, H., Liu, T.-Y. (2022). BioGPT: Generative pre-trained transformer for biomedical text generation and mining. Briefings in Bioinformatics, 23(6). arXiv:2210.10341.
- Tilton, A. K., Caplan, B. E., Cole, B. J. (2025). Generative AI in consumer health: leveraging large language models for health literacy and clinical safety with a digital health framework. Frontiers in Digital Health, 7:1616488.
- Sallam, M. (2023). ChatGPT utility in healthcare education, research, and practice: systematic review on the promising perspectives and valid concerns. Healthcare, 11(6), 887.
- Ayo-Ajibola, O., Davis, R. J., Lin, M. E., Riddell, J., Kravitz, R. L. (2024). Characterizing the adoption and experiences of users of artificial intelligence-generated health information in the United States: cross-sectional questionnaire study. Journal of Medical Internet Research, 26, e55138.
- Health consumers' use and perceptions of health information from generative artificial intelligence chatbots: a scoping review. (2025). PMC.
- Design and multidimensional expert evaluation of a national health information portal-integrated generative AI chatbot for health literacy and awareness. (2025/2026). PMC.
- Omar, M., et al. Multi-model assurance analysis showing large language models are highly vulnerable to adversarial hallucination attacks during clinical decision support. (2025). PMC.
- Harm reduction strategies for thoughtful use of large language models in the medical domain: perspectives for patients and clinicians. (2025). Journal of Medical Internet Research, 27, e75849.
- Opportunities and challenges for ChatGPT and large language models in biomedicine and health. (2023). arXiv:2306.10070.
- Privacy-by-design framework for large language model chatbots in urology. (2025/2026). PMC.
- Privacy challenges and solutions in retrieval-augmented generation-enhanced LLMs for healthcare chatbots: a review of applications, risks, and future directions. (2025). arXiv:2511.11347.
- Security and privacy challenges of large language models: a survey. (2024). arXiv:2402.00888.